

Project Proposal

Investigate Controversial Climate Change Discussions and Analyze the Key Arguments from Different Perspectives

Dataset

We propose to use the *Public Opinion on Climate Change (Updated Daily)* dataset published on Kaggle. It contains climate-related Reddit comments and associated post-, user-, temporal-, and engagement-related metadata.

It should be emphasized that this project title is based on one of the task ideas suggested by the dataset publisher on Kaggle.

In Kaggle's community feedback, the dataset has received both the “**Clean data**” and “**Original**” labels. Nevertheless, our preliminary sampling suggests that it is not completely analysis-ready. We observed missing values, repeated records or repeated post-level information, irrelevant or indirectly related comments, unstructured text, and potentially noisy content. Although the data has already been collected, filtered, combined, and converted into a CSV file, task-specific preprocessing is still required.

Research Question

What types of arguments and discussion patterns appear in controversial climate-change conversations on Reddit, and how do these patterns differ across perspectives, communities, time periods, and levels of engagement?

Reddit's controversiality field will be treated as a platform-specific indicator rather than definitive evidence of ideological controversy. We will first examine its meaning, quality, and distribution.

Rationale

Climate-change discussions may involve scientific explanations, political arguments, skepticism, conspiracy claims, personal experiences, and emotional reactions. Identifying recurring patterns may help researchers understand how climate arguments are expressed, how discussions differ across communities, and what content is associated with controversial or unusual engagement. Because Reddit users are not representative of the general population, our findings will describe patterns within Reddit rather than overall public opinion.

Possible Methods

Depending on the initial exploration and feasibility assessment, possible methods may include:

- clustering to identify naturally occurring groups of comments;
- association-rule mining to examine recurring combinations of terms, themes, communities, and engagement categories;
- anomaly detection to identify unusual comments, posts, or users;
- topic modeling to identify major discussion themes;
- sentiment analysis to compare emotional tone;
- classification, if appropriate, to investigate comments marked as controversial; and
- representative-comment analysis or argument extraction to interpret key arguments.

These methods are possibilities rather than a commitment to apply all of them. The final selection will be limited to methods that are relevant, computationally feasible, and sufficiently supported by the data.

Expected Preprocessing

Possible preprocessing includes handling missing values and duplicates, separating repeated post- and comment-level information, filtering deleted or irrelevant content, normalizing text, detecting language, transforming timestamps, constructing features, addressing class imbalance, representing text numerically, scaling metadata, reducing dimensionality, and selecting a manageable but representative sample.

Risks and Feasibility

- **Partially processed data:** Since collection and initial filtering have already been completed, the remaining preprocessing may not be considered substantial enough. We will document existing data-quality problems and distinguish previous processing from our own work.
- **Large scale:** The dataset may exceed our available memory and computational resources. We may use chunk-based processing, database-supported preprocessing, or justified sampling.
- **Ambitious scope:** Applying every proposed method may not be feasible. We will select a small number of primary methods after exploration and treat the others as optional extensions.
- **Uncertain patterns:** The controversiality indicator may be imbalanced or unrelated to meaningful differences in arguments. Any negative or inconclusive findings will be reported honestly.
- **Bias and irrelevant content:** Reddit is not representative of the general population, and the data may contain only indirectly relevant discussions.

Alternative Project Plans

Our group has also prepared other potential topics and datasets. This is our preferred option, but if its dataset, question, preprocessing requirements, or scope does not satisfy the course requirements, we are prepared to revise it or submit an alternative proposal based on the tutor's feedback.